CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 13: Section and Mid-Point Formula

Topic-wise short notes with solved examples

These formulas give the coordinates of a point that divides a line segment in a given ratio. The mid-point and the centroid of a triangle are special cases.

1 The section formula

- Let $P(x, y)$ divide the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m_1 : m_2$, so that $AP : PB = m_1 : m_2$. Then

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \qquad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

- Each part of the ratio multiplies the coordinates of the **opposite** end: m_1 goes with B and m_2 goes with A.

Example 1.1 Find the coordinates of the point that divides the segment joining $A(2, -1)$ and $B(7, 9)$ in the ratio $3 : 2$.

SOLUTION

$$x = \frac{3 \times 7 + 2 \times 2}{3 + 2} = \frac{25}{5} = 5$$
$$y = \frac{3 \times 9 + 2 \times (-1)}{3 + 2} = \frac{25}{5} = 5$$

The point is $(5, 5)$.

2 Finding the ratio

- To find the ratio in which a point divides a segment, take the ratio as $k : 1$ and use one coordinate to find k.
- A point on the x-axis has ordinate 0. A point on the y-axis has abscissa 0.
- If k comes out negative, the point divides the segment externally.

Example 2.1 In what ratio does the point $P(3, 4)$ divide the segment joining $A(1, 2)$ and $B(6, 7)$?

SOLUTION

Let the ratio be $k : 1$. Then $\frac{6k + 1}{k + 1} = 3$, so $6k + 1 = 3k + 3$ and $k = \frac{2}{3}$.

Check with y: $\frac{7k + 2}{k + 1} = \frac{\frac{14}{3} + 2}{\frac{5}{3}} = 4$. Correct.

The ratio is $2 : 3$.

Example 2.2 In what ratio does the y-axis divide the segment joining $A(-4, 5)$ and $B(6, 0)$? Find the point of intersection.

SOLUTION

Let the ratio be $k : 1$. On the y-axis the abscissa is 0: $\frac{6k - 4}{k + 1} = 0$, so $k = \frac{2}{3}$.

The ratio is $2 : 3$. Ordinate $= \frac{2 \times 0 + 3 \times 5}{2 + 3} = 3$.

Ratio $2 : 3$; the point is $(0, 3)$.

3 Points of trisection

- The points of trisection divide a segment into three equal parts. They divide it in the ratios $1 : 2$ and $2 : 1$.

Example 3.1 Find the points of trisection of the segment joining $A(2, 1)$ and $B(8, 10)$.

SOLUTION

Ratio $1 : 2$: $P = \left(\frac{8 + 4}{3}, \frac{10 + 2}{3}\right) = (4, 4)$

Ratio $2 : 1$: $Q = \left(\frac{16 + 2}{3}, \frac{20 + 1}{3}\right) = (6, 7)$

The points are $(4, 4)$ and $(6, 7)$.

4 The mid-point formula

- The mid-point of the segment joining (x_1, y_1) and (x_2, y_2) is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

- If one end and the mid-point are known, the other end can be found.
- The diagonals of a parallelogram bisect each other, so they have the same mid-point.

Example 4.1 The mid-point of the segment joining $A(3, -2)$ and B is $M(1, 4)$. Find B.

SOLUTION

Let $B = (x, y)$. Then $\frac{3 + x}{2} = 1$ and $\frac{-2 + y}{2} = 4$.

$B = (-1, 10)$

Example 4.2 $A(2,3)$, $B(6,5)$ and $C(8,9)$ are three vertices of a parallelogram $ABCD$. Find D.

SOLUTION

Mid-point of $AC = (5,6)$. This is also the mid-point of BD.

Let $D = (x,y)$: $\frac{6+x}{2} = 5$ and $\frac{5+y}{2} = 6$

$D = (4,7)$

Example 4.3 AB is a diameter of a circle with centre $C(2,-3)$. If $A = (5,1)$, find B.

SOLUTION

The centre is the mid-point of the diameter: $\frac{5+x}{2} = 2$ and $\frac{1+y}{2} = -3$.

$B = (-1,-7)$

5 Centroid of a triangle

- A **median** joins a vertex to the mid-point of the opposite side. The three medians meet at the **centroid** G.
- The centroid divides each median in the ratio $2:1$, measured from the vertex.

$$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

Example 5.1 Find the centroid of the triangle with vertices $A(1,4)$, $B(-2,-1)$ and $C(7,3)$.

SOLUTION

$G = \left(\frac{1-2+7}{3}, \frac{4-1+3}{3}\right)$

$G = (2,2)$

Example 5.2 The centroid of a triangle is $(1,3)$ and two of its vertices are $(2,5)$ and $(-4,1)$. Find the third vertex.

SOLUTION

$\frac{2-4+x}{3} = 1$ gives $x = 5$, and $\frac{5+1+y}{3} = 3$ gives $y = 3$.

The third vertex is $(5,3)$.

6 Common mistakes

- Multiplying m_1 by x_1. It multiplies x_2.
- Writing the ratio the wrong way round: $AP:PB$ is not the same as $PB:AP$.
- Forgetting that a point on the x-axis has $y = 0$, not $x = 0$.
- Using the mid-point formula with a subtraction instead of an addition.

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